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Macroprudential Regulation, Quantitative Easing, and Bank Lending

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1 Big picture

- **Question.** Do macroprudential accounting rules change how quantitative easing and other long-rate monetary policies pass through to bank lending?
- **Setting.** The paper studies Italian banks around two ECB Public Sector Purchase Programme (PSPP) announcements: January 2015 and September 2019. The key institutional change between them is accounting treatment of sovereign available-for-sale (AFS) securities.
- **Core challenge.** A naive comparison of banks with more sovereign holdings to those with fewer sovereign holdings mixes together the effect of QE, broad balance-sheet exposure, local credit demand, and banks' other business-model characteristics. The paper's core identification problem is to isolate the part of exposure that actually moves regulatory capital.
- **Main conceptual contribution.** The paper shows that what matters is not a bank's overall holdings of QE-eligible securities, but the subset valued using **mark-to-market accounting (MMA)** for regulatory purposes. **Historical cost accounting (HCA)** insulates regulatory capital from price changes and therefore mutes monetary transmission.
- **Main mechanisms.**
 - **Recapitalization channel:** QE raises the market value of eligible MMA securities, increases regulatory capital, and expands lending capacity.
 - **Portfolio-rebalancing channel:** banks may sell appreciated securities and reallocate into loans.
 - **Liquidity channel:** QE could relax funding conditions, but the paper finds little support for this as the main force.
- **Main empirical design.** The authors combine monthly Italian credit-register data, security-level supervisory holdings, and the accounting location of each sovereign asset on banks' balance sheets. They estimate bank-firm lending regressions with bank fixed effects and firm-time fixed effects around each PSPP announcement.
- **Main baseline result.** In both 2015 and 2019, only banks exposed through *MMA-valued* sovereign securities expand lending after the PSPP announcement. Broad exposure that ignores accounting treatment does not explain the lending response.
- **Main magnitude result.** The aggregate lending effect is about **EUR 12 billion** in 2019 but only about **EUR 214 million** in 2015. The gap is large because in 2015 only trading-book sovereigns were under MMA, whereas in 2019 both AFS and trading-book sovereigns were under MMA.
- **Main price result.** Loan rates fall for more exposed banks by about **60 basis points** in 2019 and about **97 basis points** in 2015, consistent with a supply shift rather than a demand shift.
- **Main mechanism result.** The data favor recapitalization over rebalancing. Capital gains on MMA-eligible securities strongly predict lending; trading activity does not, once capital gains are accounted for.

- **Allocation result.** Much of the additional credit goes to safer firms, and less-capitalized banks derisk the most after the PSPP.
- **External-validity result.** Similar logic extends beyond QE. High-frequency path-rate shocks to long yields affect lending only through banks' MMA sovereign exposure, with stronger effects for weaker-capitalized banks.
- **Economic takeaway.** HCA can stabilize bank capital against sovereign-price volatility, but it also blocks monetary policy transmission through regulatory capital. The paper therefore frames a real policy trade-off in macroprudential accounting design.

2 Data and measurement (institutional context and key objects)

2.1 Institutional background: why Italy and why accounting treatment matters

- Italian banks hold large amounts of sovereign securities, making the country a natural place to study how sovereign-bond valuation passes into bank lending.
- The paper focuses on three regulatory classifications for sovereign securities:
 - **Held to maturity (HTM):** always valued at HCA.
 - **Trading book:** always valued at MMA.
 - **Available for sale (AFS):** treated differently over time.
- Between 2010 and 2017, euro-area sovereign AFS securities were effectively valued at HCA for regulatory capital because unrealized gains and losses were filtered out.
- Starting in 2018, AFS sovereign securities were brought under MMA for regulatory purposes.

Interpretation. The same sovereign bond can matter differently for lending depending on where it sits on the balance sheet. That is the central source of economically meaningful heterogeneity in the paper.

2.2 Why the 2015 and 2019 PSPP announcements are so useful

- The first PSPP announcement occurs on January 22, 2015, when only trading-book sovereign securities are under MMA.
- The restart of the PSPP is announced on September 12, 2019, when both AFS and trading-book sovereign securities are under MMA.
- Thus, the two episodes differ sharply in how much of banks' eligible sovereign portfolios can pass through to regulatory capital.

Interpretation. The same type of monetary policy shock hits a different accounting regime. That comparison is what lets the paper identify the mediating role of macroprudential accounting.

2.3 Main data sets and sample construction

- **Italian Central Credit Register:** monthly bank-firm lending relationships; loans above EUR 30,000 are covered, so the data are near-universal.
- **AnaCredit and Taxia:** bank-firm interest rates on new term loans. AnaCredit is monthly for 2019; Taxia is quarterly for 2015.
- **Bank of Italy Supervisory Reports:** bank balance sheets and, crucially, security holdings at the ISIN level with balance-sheet classification.
- The analysis uses banking groups rather than single banks, because capital regulation is assessed at the group level.
- The final samples include about **90 banks in 2019** and **95 banks in 2015**, excluding foreign, mutual, and cooperative banks.

2.4 Main lending outcome

The baseline dependent variable is intensive-margin loan growth at the bank-firm-month level:

$$\Delta \log L_{bft} = \log(L_{bft}) - \log(L_{bf,t-1}), \quad (1)$$

where L_{bft} is the amount disbursed by bank b to firm f in month t .

Interpretation. The paper is asking whether the same firm borrows more from a more exposed bank, holding firm-month demand fixed.

2.5 Interest-rate outcome

For newly originated term loans, the paper studies the interest rate i_{bft} on the new loan.

- The 2019 interest-rate sample contains about **14,574** observations.
- The 2015 sample contains about **9,451** observations.

Interpretation. Rates are used as a complementary supply-side check: if quantities rise and prices fall, the case for loan-supply expansion is much stronger.

2.6 Exposure to QE: broad versus MMA exposure

The broad exposure measure is the ratio of PSPP-eligible securities to total assets:

$$QE_b^{broad} = \frac{\text{PSPP-eligible securities}_b}{\text{Assets}_b}. \quad (2)$$

The more informative measure uses only PSPP-eligible securities that are valued at MMA:

$$QE_b^{MMA} = \frac{\text{PSPP-eligible securities under MMA}_b}{\text{Assets}_b}. \quad (3)$$

Interpretation.

- The broad measure captures total potential exposure to ECB purchases.
- The MMA measure captures the part of that exposure that actually changes regulatory capital when yields move.

2.7 Capital gains and trading activity

To distinguish recapitalization from portfolio rebalancing, the paper constructs two bank-level variables in the announcement month.

Capital gains on MMA-eligible securities are:

$$\text{CapitalGains}_{b,t} = \frac{\sum_s \text{Quantity}_{s,b,t-1} (\text{Price}_{s,t} - \text{Price}_{s,t-1})}{\text{Assets}_{b,t-1}}. \quad (4)$$

Trading revenues are:

$$\text{Trading}_{b,t} = \frac{\sum_s (\text{Quantity}_{s,b,t} - \text{Quantity}_{s,b,t-1}) \cdot \text{Price}_{s,t}^{avg}}{\text{Assets}_{b,t-1}}. \quad (5)$$

Interpretation.

- $\text{CapitalGains}_{b,t}$ is the recapitalization-channel object.
- $\text{Trading}_{b,t}$ is the rebalancing-channel object.

2.8 Bank capitalization, liquidity, and firm-risk measures

- **Tier 1 ratio:** Tier 1 capital over risk-weighted assets, measured at the previous year-end.
- **Liquidity:** central-bank deposits plus cash, over assets.
- **Risky firm dummy:** equals 1 for firms with credit-risk score at least 7 on a 0–10 scale.
- **Big firm dummy:** equals 1 if revenues exceed EUR 50 million.

Interpretation. These heterogeneity variables let the paper test whether the monetary transmission channel works through regulatory capital and how that channel shapes the allocation of credit.

2.9 Descriptive facts

- On average, banks hold PSPP-eligible securities equal to **17.06% of assets** in August 2019 and **11.44%** in December 2014.
- Eligible sovereign securities under MMA average **6.00% of assets** in 2019 but only **0.45%** in 2015.
- Eligible securities under HCA average about **11.12% of assets** in 2019 and **11.00%** in 2015.
- In the lending sample with multiple bank relationships, there are more than **8 million** bank-firm-month observations around 2019 and more than **6 million** around 2015.

Interpretation. The key fact is not just that banks hold many sovereigns, but that the part under MMA changes enormously between 2015 and 2019.

3 Models and empirical specifications (all equations + interpretation)

3.1 Mechanism and hypotheses

The paper’s basic economic logic is:

- QE or other long-rate easing raises the price of long-duration sovereign bonds.
- If those bonds are under MMA, the price increase passes into regulatory capital.
- If capital constraints matter for lending, more exposed banks expand credit supply.
- If the securities are under HCA, the price change does *not* move regulatory capital, so the lending channel is muted.

This gives three concrete empirical predictions:

1. lending should respond to *MMA exposure*, not broad exposure;
2. weaker-capitalized banks should react more;
3. recapitalization should matter more than pure trading or liquidity.

3.2 Baseline event-study specification around the 2019 announcement

The main 2019 lending specification is:

$$\Delta \log L_{bft} = \sum_{\tau \neq \text{Aug } 2019} \beta_{\tau} I_{\tau} \times QE_b + \sum_{\tau \neq \text{Aug } 2019} \gamma_{\tau} I_{\tau} \times Y_b + \delta Z_{b,t} + \psi_b + \psi_{f,t} + \varepsilon_{bft}. \quad (6)$$

Here:

- I_τ is a month dummy in the 12-month event window,
- QE_b is the bank's pre-announcement exposure measure,
- Y_b contains bank size and central-bank reserves interacted with month dummies,
- $Z_{b,t}$ is the change in ECB borrowing,
- ψ_b are bank fixed effects,
- $\psi_{f,t}$ are firm-time fixed effects.

Interpretation. The coefficient sequence β_τ asks whether, in the same firm-month, banks with higher pre-announcement exposure expand lending more after the policy shock.

3.3 2015 specification

The 2015 specification is the same design centered around January 2015, with exposure measured as of December 2014.

Interpretation. The comparison of 2015 and 2019 is not merely before-versus-after QE. It is the same style of QE shock passing through two different accounting environments.

3.4 Difference-in-difference summary version

For exposition, the paper also estimates a concise version:

$$\Delta \log L_{bft} = \alpha + \beta Post_t \times QE_b + \gamma Y_b + \delta Z_{b,t} + \psi_b + \psi_{f,t} + \varepsilon_{bft}, \quad (7)$$

where $Post_t$ equals 1 in the announcement month and the following months.

Interpretation. This version summarizes the average post-announcement effect and is used in several formal comparisons.

3.5 Formal 2015–2019 test of the role of MMA

The paper nests the two windows and tests whether AFS exposure matters only when AFS securities are under MMA:

$$\Delta \log L_{bft} = \theta Post_t \times QE_b^{AFS} + \lambda Post_t \times QE_b^{AFS} \times I_{2019} + \beta Post_t \times QE_b^{TradingBook} + \text{controls}. \quad (8)$$

Interpretation.

- θ captures the AFS effect when AFS is effectively under HCA (the 2015 regime).
- λ captures the extra AFS effect when AFS is under MMA (the 2019 regime).

A positive and significant λ together with an insignificant θ is exactly what the accounting-mediated transmission story predicts.

3.6 Interest-rate regression

Using the smaller sample of new term loans, the paper estimates:

$$i_{bft} = \alpha + \beta \text{Announcement}_t \times QE_b + \text{controls} + \psi_b + \psi_{f,t} + u_{bft}. \quad (9)$$

Interpretation. If more exposed banks lower rates after the announcement, then the lending results reflect loan-supply shifts rather than demand increases.

3.7 Recapitalization versus liquidity

To compare channels, the paper interacts exposure with bank capital and liquidity:

$$\Delta \log L_{bft} = \alpha + \beta_1 \text{Ann}_t \times QE_b + \beta_2 \text{Ann}_t \times QE_b \times \text{Tier1}_b + \beta_3 \text{Ann}_t \times QE_b \times \text{Liquidity}_b + \dots \quad (10)$$

Interpretation.

- A negative β_2 means less-capitalized banks react more strongly, supporting the recapitalization channel.
- A large role for β_3 would support a liquidity channel.

3.8 Trading specification at the security level

The trading regression uses security-by-bank-by-month observations:

$$\text{Trading}_{sbt} = \sum_{\tau} \beta_{\tau} I_{\tau} \times \text{MMAExposure}_b + \sum_{\tau} \gamma_{\tau} I_{\tau} \times \text{HCAExposure}_b + \psi_b + \psi_{s,t} + u_{sbt}. \quad (11)$$

Interpretation. This design asks whether banks with more MMA exposure sell eligible AFS securities after PSPP announcements, which would be the first step in a portfolio-rebalancing story.

3.9 Horse race between capital gains and trading

The paper then runs a direct horse race:

$$\Delta \log L_{bft} = \alpha + \beta \text{Ann}_t \times \text{Trading}_{b,t} + \kappa \text{Ann}_t \times \text{CapitalGains}_{b,t} + \text{controls}. \quad (12)$$

Interpretation. If trading matters only because it realizes capital gains, then κ should survive while β should not.

3.10 Allocation-of-credit specification

To study risk taking, the paper interacts exposure with firm risk and size:

$$\Delta \log L_{bft} = \alpha + \beta_1 Ann_t \times QE_b + \beta_2 Ann_t \times QE_b \times Risky_f + \beta_3 Ann_t \times QE_b \times Big_f + \dots \quad (13)$$

It also adds a quadruple interaction with bank capital:

$$\Delta \log L_{bft} = \dots + \eta Ann_t \times QE_b \times Risky_f \times Tier1_b + \dots \quad (14)$$

Interpretation. These coefficients show whether the new lending tilts toward safer or riskier borrowers and whether undercapitalized banks derisk more.

3.11 Path-rate-shock specification for broader monetary transmission

To go beyond the PSPP, the paper uses high-frequency path-rate shocks to 10-year Italian yields from the EA-MPD database:

$$\Delta \log L_{bft} = \beta IT10_t \times \left(\frac{Sovereign MMA_{5+}}{Assets} \right)_{b,t-1} + \gamma Y_{b,t-1} + \delta Z_{b,t} + \psi_b + \psi_{f,t} + \varepsilon_{bft}. \quad (15)$$

Interpretation. A negative β means that a negative long-rate shock raises lending more for banks with greater long-duration MMA sovereign exposure.

3.12 Simple discussion framework

In the paper's stripped-down capital-constraint discussion, bank capital is

$$capital_t = l_{t-1} + p \cdot g - d_{t-1}, \quad (16)$$

and the bank must satisfy

$$capital_t \geq \xi l_t. \quad (17)$$

If sovereigns are under HCA, then $p = p_{t-1}$ and current price changes do not affect regulatory capital. If they are under MMA, then $p = p_t$ and yield changes pass through immediately.

The paper also proposes a hybrid rule that lets the risk-free component pass through but freezes the sovereign spread at its historical value. In reduced form, that preserves monetary-policy transmission while muting sovereign-spread shocks.

Interpretation. This section is not the source of causal identification. It is the paper's way of clarifying the policy trade-off revealed by the empirical results.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the design is unusually clean

Three features make the design especially convincing:

- the policy shocks are sharp announcement events,
- the event windows are narrow,
- and the data are monthly at the bank-firm-security level.

This helps isolate QE from other contemporaneous ECB policies such as negative rates, forward guidance, and targeted refinancing operations.

4.2 Loan demand is absorbed nonparametrically

The firm-time fixed effects $\psi_{f,t}$ are a major strength of the design.

- They absorb all firm-specific loan demand shocks in a given month.
- Identification therefore comes from comparing how different banks change lending to the *same firm* in the *same month*.

Why this matters. A lot of bank-lending papers struggle to separate supply from demand. This one attacks that problem directly.

4.3 Granular balance-sheet data are essential

A coarse exposure measure would mix together assets that are treated very differently by regulation. The paper uses security-level holdings and balance-sheet location to know whether each eligible sovereign security is under MMA or HCA.

Why this matters. Without that classification, the core mechanism would be unobservable and broad exposure would be contaminated by measurement error.

4.4 Why broad exposure fails but MMA exposure works

This is not just a descriptive quirk. It is an identification result.

- Broad exposure is not predictive of lending after PSPP announcements.
- MMA exposure is predictive.
- AFS exposure matters in 2019, when AFS is under MMA, but not in 2015, when AFS is effectively under HCA.

Interpretation. The paper is not merely showing that banks with more sovereigns lend more after QE. It is showing that the *regulatory valuation regime* determines whether sovereign holdings transmit the policy shock.

4.5 Reverse causality is weak

The lending regressions use pre-announcement exposure and focus on immediate post-announcement movements in lending. It is therefore difficult to argue that expected lending changes determined banks' prior classification of sovereign assets in a way that lines up with the timing of PSPP announcements.

4.6 Pre-trends and anticipation

The event-study coefficients show no systematic pre-trends for the MMA exposure specifications.

- For 2019, the average March–July pre-period coefficient is about **0.04** with a p-value of 0.48 in the MMA specification.
- For 2015, the average July–November 2014 pre-period coefficient is about **0.13** with a p-value of 0.50.

The paper also argues that accounting classifications are sticky and difficult to relabel opportunistically because of supervisory scrutiny.

4.7 Alternative explanations confronted in the paper

The paper rules out several alternatives:

- **Credit demand:** loan rates fall where loan quantities rise.
- **Generic bank strength:** the 2015 versus 2019 comparison changes which balance-sheet items are relevant, helping separate accounting treatment from fixed bank traits.
- **Liquidity channel:** little support once capital and liquidity are separately interacted.
- **Pure trading channel:** trading rises for AFS securities in 2019, but does not explain lending once capital gains are included.
- **Other ECB programs and bank characteristics:** the Internet Appendix adds many additional controls and the core result survives.

4.8 Limits of the design

- The paper is strongest on short-run loan-supply responses around clean policy announcements.
- The broader path-rate-shock analysis is more suggestive than the PSPP event studies because it uses a monthly shock series rather than isolated announcement windows.

- The evidence is for Italy, which is an especially suitable but not necessarily universal setting because sovereign holdings are large and the sovereign-bank loop is salient.

4.9 Relation to the literature

The paper links three literatures that are usually separate:

- QE and the bank lending channel,
- macroprudential policy and capital regulation,
- accounting treatment of financial assets.

Its distinctive claim is that accounting-based macroprudential rules directly mediate the real effects of monetary policy.

5 Tables and figures

5.1 Figure 1 and Figure 2: the institutional timeline and the accounting map

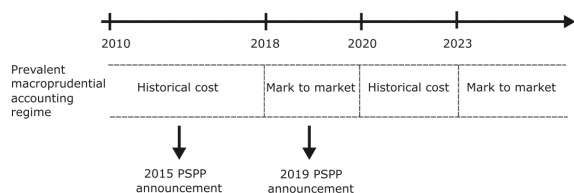


Figure 1
Macroeprudential accounting regime of sovereign securities and QE announcements
The figure shows the evolution of some key aspects of the European Union macroprudential accounting regime that banks use to value their holdings of sovereign securities when computing regulatory net worth (see Section 2 for more details) and the two main announcements of the PSPP, which is the ECB's largest QE program.

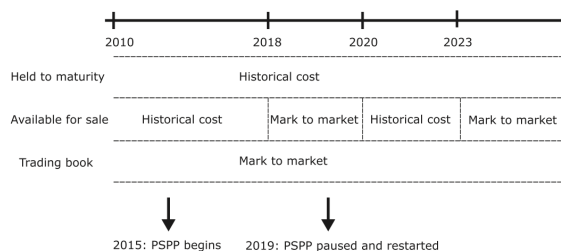


Figure 2
Sovereign security classification and prevalent macroprudential accounting regime
This figure shows how identical sovereign assets on the balance sheet are valued differently depending on their regulatory classification, for the purpose of computing regulatory capital. Between 2010 and 2018, sovereign assets classified as available for sale were valued at historical cost. Between 2018 and 2019, sovereign assets classified as available for sale were then marked to market. In 2020, amid the COVID-19 pandemic, regulators introduced a *prudential filter* that allowed banks to again value available-for-sale sovereign assets at historical cost, based on the market value as of December 31, 2019. Assets classified as held to maturity are always valued at historical cost, whereas those in the trading book are always marked to market.

Read:

- Figure 1 lays out the evolution of the macroprudential accounting regime and places the two main PSPP announcements on that timeline.
- Figure 2 shows the accounting classification logic: HTM always HCA, trading book always MMA, and AFS sovereigns moving from effective HCA to MMA in 2018.

Economic message. These figures are the institutional heart of the paper. They explain why the same QE-style intervention should transmit very differently in 2015 and 2019.

5.2 Table 1 and Table 2: what the sample looks like

Table 1
Summary statistics, firm-bank lending relationships

A. Number of lending relationships

	August 2019		December 2014	
	N	Percentage	N	Percentage
2	177,719	60	141,323	65
3	62,810	21	44,983	21
4	26,928	9	17,089	8
5	13,101	4	7,330	3
6+	17,331	6	5,993	3
Total	297,889	100	216,718	100

B. Firm-bank lending relationships

Firm-bank lending relationships	N	Mean	1st quartile	Median	3rd quartile	Std. dev.
<i>All firms, 2019 PSPP</i>						
Amount borrowed, EUR (as of August 2019)	1,628,965	368,983	32,474	71,806	192,770	5,241,722
log change amount borrowed, % (12-month window)	17,130,567	-0.56	-3.33	-0.33	0.38	50.56
<i>Firms with more than one bank relationship in each month, 2019 PSPP</i>						
Amount borrowed, EUR (as of August 2019)	846,286	533,090	33,061	93,089	287,670	6,942,719
log change amount borrowed, % (12-month window)	8,780,566	-0.77	-5.72	-0.32	1.99	57.20
<i>All firms, 2015 PSPP</i>						
Amount borrowed, EUR (as of December 2014)	1,216,779	414,973	36,370	79,218	211,910	5,444,324
log change amount borrowed, % (12-month window)	13,186,871	0.17	-2.04	0.00	0.61	46.33
<i>Firms with more than one bank relationship in each month, 2015 PSPP</i>						
Amount borrowed, EUR (as of December 2014)	563,922	625,108	43,978	115,779	346,191	7,587,313
log change amount borrowed, % (12-month window)	6,117,128	0.19	-2.97	0.00	1.24	51.16

C. Interest rates on new term loans

Interest rates on new term loans	N	Mean	1st quartile	Median	3rd quartile	Std. dev.
<i>All observations, 2019 PSPP</i>						
Interest rate, % (12-month window)	248,439	4.18	2.44	3.80	5.52	2.35
<i>Firms with new term loans from multiple banks in each month, 2019 PSPP</i>						
Interest rate, % (12-month window)	14,580	3.17	1.63	2.67	4.18	2.10
<i>All observations, 2015 PSPP</i>						
Interest rate, % (12-month window)	127,369	4.82	3.07	4.35	6.13	2.46
<i>Firms with new term loans from multiple banks in each quarter, 2015 PSPP</i>						
Interest rate, % (12-month window)	9,451	3.97	2.39	3.54	5.03	2.19

Panel A displays the distribution of the number of lending relationships for the firms with outstanding loans reported in the Italian credit register. Panel B displays the distribution of the amount borrowed and the log change in the amount borrowed. Panel C displays the interest rate distribution for new term loans.

Table 2
Bank-level summary statistics

		Mean	1st quartile	Median	3rd quartile	Std. dev.	Coeff. var.
Eligible securities, % of assets	Aug 2019	17.06	7.14	15.14	24.42	13.39	0.79
	Dec 2014	11.44	1.69	8.10	18.37	11.35	0.99
Eligible securities, marked to market, % of assets	Aug 2019	6.00	0.18	3.16	6.66	9.74	1.62
	Dec 2014	0.45	0.00	0.00	0.001	1.63	1.63
Eligible securities, historical cost, % of assets	Aug 2019	11.12	3.09	7.68	18.18	10.41	0.94
	Dec 2014	11.00	1.31	8.10	17.46	11.18	1.02
Eligible securities, available for sale and trading book, % of assets	Aug 2019	6.00	0.18	3.16	6.66	9.74	1.62
	Dec 2014	10.42	0.17	6.69	17.65	11.02	1.06
Noneligible securities, available for sale, % of assets	Aug 2019	3.65	0.29	1.36	4.69	5.57	1.53
	Dec 2014	9.01	1.47	5.66	12.68	10.64	1.18
Noneligible securities, trading book, % of assets	Aug 2019	4.81	0.15	2.24	6.59	7.11	1.48
	Dec 2014	4.12	0.00	0.19	6.30	6.46	1.57
log of total assets	Aug 2019	21.85	20.25	21.53	23.39	2.16	0.10
	Dec 2014	21.38	19.41	21.45	22.82	2.20	0.10
Business loans, % of assets	Aug 2019	25.57	7.88	22.72	32.11	38.39	1.51
	Dec 2014	25.24	10.85	26.75	37.71	15.59	0.62
Liquidity (i.e., cash and central bank deposits), % of assets	Aug 2019	3.25	0.27	1.19	3.11	6.67	2.05
	Dec 2014	1.31	0.08	0.76	1.29	1.82	1.39
Tier 1 capital, % of risk-weighted assets	Aug 2019	19.14	12.34	15.17	19.05	12.58	0.66
	Dec 2014	22.10	10.60	14.18	26.00	24.56	1.11
TLTRO exposures, % of total assets	Feb 2019	9.52	5.83	9.98	12.60	5.17	0.54
	Dec 2014	0.33	0.00	0.00	1.00	2.30	6.65
Net interbank position, % of assets	Jun 2019	4.08	-0.64	0.57	4.29	11.23	2.75
	Mar 2014	-0.35	-3.73	0.00	2.97	12.32	-35.64
Securitization dummy	Aug 2019	0.48	0.00	0.00	1.00	0.50	1.05
	Aug 2014	0.28	0.00	0.00	1.00	0.45	1.60
Covered bonds and ABS, available for sale and trading book, % of assets	Aug 2019	1.64	0.00	0.00	1.33	3.58	2.19
	Aug 2014	1.68	0.00	0.00	1.53	3.51	2.09
CSPP-eligible securities, available for sale and trading book, % of assets	Aug 2019	0.09	0.00	0.00	0.06	0.25	2.65
	Aug 2014	0.09	0.00	0.00	0.06	0.25	2.65
Unused allowance, two-tier reserve system, % of assets	Jun to Sept 2019	0.80	0.00	0.00	0.91	1.68	2.12
	Mar 2019 to Feb 2020	-2.64	0.00	0.00	0.00	183.64	-49.54
log change of borrowing from the ECB, % of assets	Jul 2014 to Jun 2015	-12.11	0.00	0.00	0.00	268.87	-22.21

The table displays key summary statistics for the banks in our sample. We have 95 banks in 2014–2015 and 90 in 2019–2020. Tier 1 data are available for 89 banks in 2019–2020 and 93 banks in 2014–2015. Cash is from end-of-year balance sheets, and thus, measured at year-end prior to each announcement. See [Internet Appendix B](#) for the definition of TLTRO exposure, net interbank position, securitization dummy, and unused allowance.

Read:

- Table 1 shows near-universal lending data, many bank relationships per firm, and large bank-firm panels in both episodes.
- In the multiple-bank subsample, there are over 8.7 million observations around 2019 and over 6.1 million around 2015.
- Table 2 shows that banks hold a lot of eligible sovereign debt, but the key change is in the amount under MMA: 6.00% of assets in 2019 versus only 0.45% in 2015.

Economic message. The difference in *MMA exposure*, not just total sovereign holdings, is what makes the two episodes economically distinct.

5.3 Table 3, Figure 3, and Figure 4: the baseline lending response

Read:

- In 2019, broad exposure does not explain loan growth, but MMA exposure does.
- Using the continuous MMA measure, the key coefficients are positive in September 2019 and December 2019; using the dummy measure, top-tercile exposed banks raise loan growth by about **2.61 percentage points** on impact and **3.37 percentage points** in December.

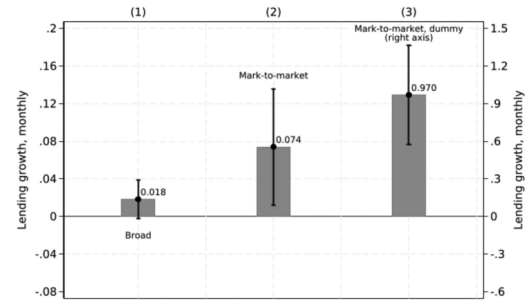
Table 3

The impact of the PSPP announcement on bank lending

	September 2019 announcement				January 2015 announcement			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
[Month -6] $\times QE_b$	-0.008 [0.015]	0.021 [0.078]	0.845 [0.970]		-0.050 [0.040]	0.079 [0.133]	0.362 [0.633]	
[Month -5] $\times QE_b$	0.147*** [0.036]	0.106 [0.161]	-0.197 [1.739]		-0.156*** [0.056]	0.183 [0.180]	0.439 [0.635]	
[Month -4] $\times QE_b$	0.054** [0.025]	0.084 [0.068]	0.833 [0.689]		-0.003 [0.055]	0.278 [0.234]	0.836 [0.582]	
[Month -3] $\times QE_b$	0.004 [0.024]	0.042 [0.125]	1.349 [1.554]		-0.089 [0.055]	-0.105 [0.243]	-0.910 [0.938]	
[Month -2] $\times QE_b$	0.115*** [0.033]	0.045 [0.149]	0.028 [1.715]		-0.010 [0.063]	0.232 [0.284]	-0.363 [0.925]	
Announcement month $\times QE_b$	0.111** [0.047]	0.244** [0.116]	2.610** [1.038]		-0.080 [0.054]	0.437** [0.182]	1.759** [0.733]	
[Month +1] $\times QE_b$	0.127** [0.061]	-0.071 [0.152]	1.136 [1.501]		-0.078** [0.039]	0.139 [0.187]	0.022 [0.703]	
[Month +2] $\times QE_b$	-0.016 [0.029]	0.111 [0.114]	1.928 [1.501]		0.009 [0.055]	0.058 [0.171]	0.368 [0.558]	
[Month +3] $\times QE_b$	0.104** [0.040]	0.339*** [0.107]	3.367*** [0.700]		-0.085* [0.045]	0.084 [0.172]	0.022 [0.569]	
[Month +4] $\times QE_b$	0.111*** [0.035]	0.071 [0.146]	-0.455 [1.758]		-0.119** [0.051]	0.039 [0.194]	-0.563 [0.766]	
[Month +5] $\times QE_b$	-0.024 [0.015]	0.031 [0.051]	-0.161 [0.379]		0.007 [0.055]	0.055 [0.138]	0.207 [0.504]	
[Post] $\times QE_b$				0.074** [0.037]				0.080* [0.048]
Eligible securities in QE_b	All	MMA only	MMA only (dummy)	MMA only	All	MMA only	MMA only (dummy)	MMA only
Bank FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm-time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Size	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ECB lending	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Reserves	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,346,925	8,346,925	8,346,925	8,346,925	5,867,308	5,867,308	5,867,308	5,670,501
R-squared	.370	.370	.370	.369	.394	.394	.394	.395

The dependent variable is the change in the logarithm of disbursed loans from bank b to firm f in month t (in percentages). Columns 1–4 refer to the 2019 PSPP announcement, and columns 5–8 to the 2015 announcement. In columns 1 and 5 the QE exposure variable, $Q E_b$, does not distinguish between MMA and HCA accounting. Columns 2 and 6 measure $Q E_b$ using only MMA assets. Columns 3 and 7 use a dummy measure of MMA exposure. Column 3 compares banks in the top tercile of the MMA exposure distribution with those in the bottom tercile, and column 7 compares banks in the top 15% with those in the bottom 85%. Columns 4 and 8 use a “post” indicator that equals 1 in the announcement month and the following months. In column 8, the sample is restricted to bank-firm relationships present in December 2014 and January 2015. Standard errors are clustered at the bank level. Boldface corresponds to the month in which the policy is announced. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A 2019 announcement



B 2015 announcement

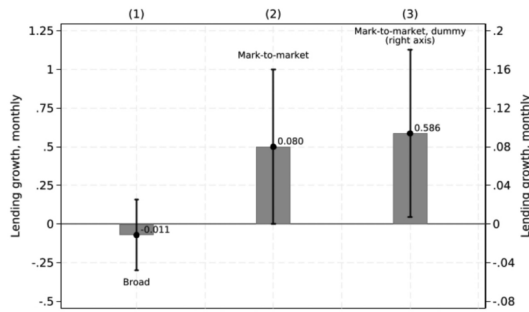


Figure 3

The impact of the PSPP announcements, post indicator

Panel A replicates columns 1–3 of Table 3, and panel B replicates columns 5–7 of Table 3, using a post indicator variable interacted with the exposure measures. The figure plots the point estimates and 90% confidence intervals. The specification in (1) uses the broad measure of exposure (i.e., all eligible securities) as a fraction of total assets (in percentages), (2) uses the MMA exposure as a fraction of total assets (in percentages), and (3) uses the MMA exposure dummy described in Section 4.1. All the regressions include controls for size, ECB lending, and central bank reserves, as well as bank and firm-time fixed effects. In panel B, the sample is restricted to bank-firm relationships present in December 2014 and January 2015. Standard errors are clustered at the bank level.

- In 2015, broad exposure again does not matter, but MMA exposure based on trading-book holdings does. With the dummy measure, top-15%-exposed banks raise loan growth by about **1.76 percentage points** in January 2015.
- The aggregate lending effect is about **EUR 12 billion** in 2019 versus only **EUR 214 million** in 2015.
- Figure 4 shows that the difference is large across the entire exposure distribution, not only at the mean.

Economic message. QE works through the part of sovereign portfolios that actually moves regulatory capital. Once that point is visible, the 2015–2019 contrast becomes economically transparent.

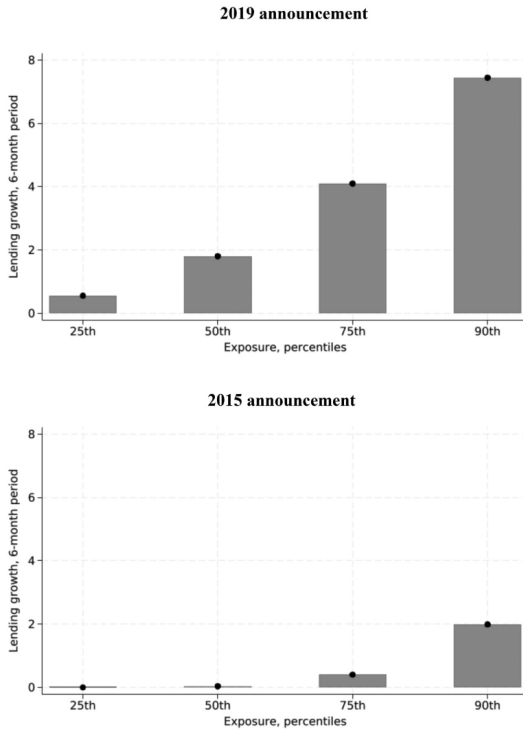


Figure 4
Effect of the PSPP announcement on bank loan supply
The figure plots the impact of the PSPP on bank lending over the 6-month period after the announcement (i.e., September 2019 to February 2020 for the 2019 announcement, and January to June 2015 for the 2015 announcement), as a function of a bank's mark-to-market exposure for banks with a positive exposure.

Table 4
Combining the 2019 and 2015 samples: A formal test of the role of MMA

	(1)	(2)	(3)	(4)
$[post] \times QE_b^{AFS}$	-0.017 [0.024]	-0.002 [0.023]	-0.007 [0.023]	
$[post] \times QE_b^{AFS} \times [I_{2019}]$	0.095** [0.041]	0.081* [0.043]	0.081* [0.041]	
$[Announcement\ month] \times QE_b^{AFS}$				0.021 [0.037]
$[Announcement\ month] \times QE_b^{AFS} \times [I_{2019}]$				0.148** [0.074]
Bank-announcement FEs	Yes	Yes	Yes	Yes
Firm-time FEs	Yes	Yes	Yes	Yes
Size, ECB lending, reserves	Yes	Yes	Yes	Yes
Size $\times [I_{2019}]$, ECB lending $\times [I_{2019}]$, reserves $\times [I_{2019}]$		Yes	Yes	Yes
$[post] \times QE_b^{TradingBook} \times [I_{2019}]$			Yes	Yes
Observations	14,214,233	14,214,233	14,214,233	14,214,233
R-squared	.378	.378	.378	.378

The dependent variable is the change in the logarithm of disbursed loans from bank b to firm f in month t (in percentages). The sample includes all observations in the 12 months around each of the 2015 and 2019 announcements. $Post$ is dummy equal to one for each of the announcement months and the following months (i.e., September 2019 to February 2020, and January to June 2015), and $Announcement\ month$ is a dummy equal to one for each of the announcement months (i.e., September 2019 and January 2015). QE_b^{AFS} denotes the holdings of eligible available-for-sale securities of bank b relative to total assets (in percentages), in the preannouncement month, and $[I_{2019}]$ is a dummy equal to 1 for the observations in the 12 months around the 2019 announcement. Column 1 uses the baseline controls of Equation (1); columns 2–4 add the same set of controls interacted with the $[I_{2019}]$ dummy; and columns 3 and 4 add the trading book exposure $QE_b^{TradingBook}$ interacted with monthly dummies and $[I_{2019}]$. Standard errors are clustered at the bank level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.4 Table 4: the formal MMA test

Read:

- The coefficient on $Post \times QE^{AFS} \times I_{2019}$ is positive and significant, around **0.095** in the baseline nested specification.
- The coefficient on $Post \times QE^{AFS}$ alone is near zero and insignificant.
- In the announcement-month-only design, the interaction rises to about **0.148**, again significant.

Economic message. AFS sovereigns matter for lending only when they are under MMA. This is the cleanest formal test in the paper of the mediating role of accounting rules.

5.5 Table 5: interest rates fall where quantities rise

Read:

- Exposed banks lower new-loan rates by about **59.8 basis points** in the 2019 announcement month.
- In 2015 the rate decline is about **97.1 basis points** at the quarterly frequency.

Economic message. The combination of higher quantities and lower prices is hard to reconcile with a pure demand story. The PSPP shifts credit supply.

Table 5
The impact of the PSPP on interest rates, 2019 and 2015

	2019 announcement (1)	2015 announcement (2)
$[2019m9] \times QE_b$	-0.598** [0.238]	$[2015q1] \times QE_b$ -0.971*** [0.351]
QE_b measure	Dummy	Dummy
Bank FEs	Yes	Yes
Firm-time FEs	Yes	Yes
Size, ECB lending, and reserves	Yes	Yes
Observations	14,574	9,451
R-squared	.807	.791

The dependent variable is the interest rate on new term loans by bank b to firm f in month t (column 1) or quarter t (column 2; both in percentages). In column 1, the variable QE_b is a dummy equal to 1 if a bank is in the top tercile of the distribution of mark-to-market eligible securities over assets as of August 2019, and the results are reported relative to banks in the bottom tercile. In column 2, QE_b is a dummy equal to 1 if a bank is in the top 15% of the distribution of mark-to-market eligible securities over assets as of December 2014. Standard errors are clustered at the bank level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6
Mechanism: Capital and liquidity

	2019 announcement (1)	2015 announcement (2)
$[2019m9] \times QE_b$	0.204** [0.101]	$[2015m1] \times QE_b$ 0.692** [0.286]
$[2019m9] \times QE_b \times [\text{Tier 1 ratio}]_b$	-0.010** [0.005]	$[2015m1] \times QE_b \times [\text{Tier 1 ratio}]_b$ -0.162** [0.069]
$[2019m9] \times QE_b \times [\text{liquidity/assets}]_b$	-0.009 [0.021]	$[2015m1] \times QE_b \times [\text{liquidity/assets}]_b$ 0.049 [0.434]
$[2019m9] \times QE_b \times [\text{Tier 1 ratio}]_b \times [\text{liquidity/assets}]_b$	-0.003 [0.004]	$[2015m1] \times QE_b \times [\text{Tier 1 ratio}]_b \times [\text{liquidity/assets}]_b$ 0.085 [0.088]
Bank FEs	Yes	Yes
Firm-time FEs	Yes	Yes
Size, ECB lending, and reserves	Yes	Yes
Observations	8,295,389	5,623,310
R-squared	.370	.396

The dependent variable is the change in the logarithm of disbursed loans from bank b to firm f in month t (in percentages). The variable QE_b is the ratio of mark-to-market eligible securities to total assets as of August 2019 (column 1) or as of December 2014 (column 2, both in percentages). The Tier 1 ratio is the ratio of Tier 1 capital to risk-weighted assets (in percentages), and liquidity denotes central bank deposits and cash over total assets (in percentages). Tier 1 ratio and liquidity/assets are centered on their respective first quartile levels (i.e., for capital, we subtract the first quartile level of the Tier 1 capital ratio from each bank's Tier 1 ratio, and similar for liquidity). All regressions include the lower-order terms unless they are absorbed by the fixed effects. Standard errors are clustered at the bank level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.6 Table 6: recapitalization versus liquidity

Read:

- The interaction between PSPP exposure and the Tier 1 ratio is negative and significant in both 2019 and 2015.
- For a median-exposure bank in 2019, lending rises about **0.83 percentage points** at the 25th percentile of the Tier 1 ratio but only about **0.57 percentage points** at the 75th percentile.
- In 2015, the contrast is even sharper: roughly **0.6 percentage points** at the 25th percentile and essentially zero at the 75th percentile.
- Liquidity interactions are economically small and statistically unimportant.

Economic message. The mechanism runs through regulatory capital, not simply through liquidity relief.

5.7 Figure 5, Figure 6, Figure 7, Table 7, and Figure 8: trading versus capital gains

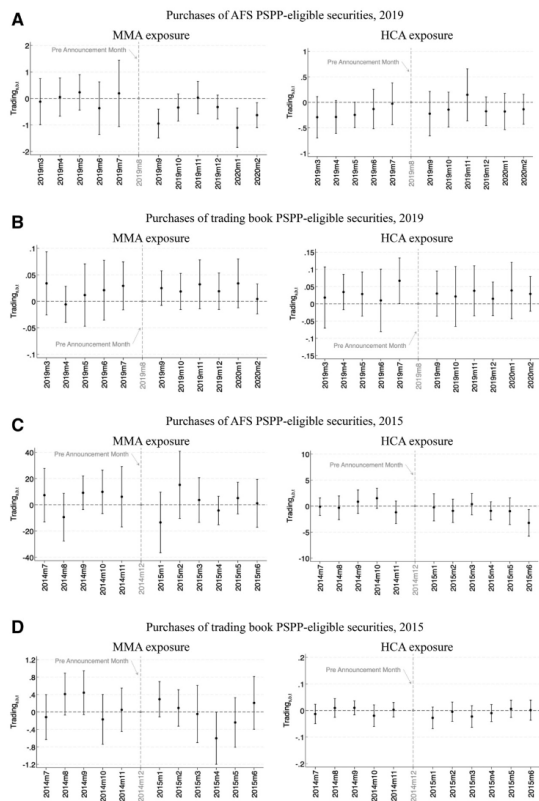


Figure 5
The impact of the PSPP on bank trading activity
The figure presents the result of a series of regressions in which the dependent variable is the purchase in EUR (or sale if negative) of security s of bank b in month t per EUR millions of total assets in $t-1$, where a security is defined at the ISIN level. Panels A and B use a 12-month window around the 2019 announcement. Panel A refers to one single specification in which s is a PSPP-eligible security in the AFS portfolio and the specification includes two measures of exposure: one based on MMA PSPP-eligible securities and the other based on HCA PSPP-eligible securities. Panel B refers to one single specification in which s is a PSPP-eligible security in the trading book, and the specification includes two measures of exposure: one based on MMA PSPP-eligible securities and the other based on HCA PSPP-eligible securities. Panels C and D follow an identical pattern for the 2015 announcement. All regressions include bank fixed effects, security-time fixed effects, and the baseline controls (i.e., size, ECB lending, and reserves). Standard errors are clustered at the bank and security level. The bars represent the 90% confidence interval around each point estimate.

Read:

- Figure 5 shows a trading response only for AFS eligible securities in 2019, exactly where regulatory friction to selling is lowest. There is no corresponding AFS response in 2015, and no important trading-book response in either year.
- In 2019, a bank at the 75th percentile of MMA exposure sells about **7.8% more** of its MMA-eligible securities than a bank at the 25th percentile.
- Table 7 shows that a one-standard-deviation increase in capital gains raises lending by about **2.36 percentage points** in 2019 and about **0.45 percentage points** in 2015.
- The coefficient on trading activity is insignificant in both episodes.
- Figure 8 shows that the effect of capital gains is concentrated in the announcement month itself.

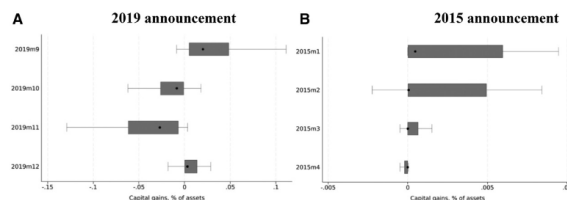


Figure 6
Evolution of capital gains after the PSPP announcement
The figure presents box plots of the evolution of capital gains in the month of the PSPP announcement and in the following 3 months, for banks with strictly positive holdings of MMA-eligible securities in the preannouncement month. We plot the median (black dot), the 25th/75th percentiles (lower/upper ends of box), and the lower/upper adjacent values (lower/upper limits on whiskers). The lower (upper) adjacent value is the furthest observation within one and a half interquartile ranges of the lower (upper) end of the box.

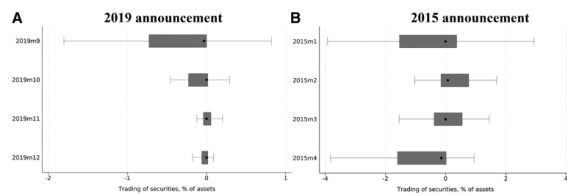


Figure 7
Evolution of trading after the PSPP announcement
The figure presents box plots of the evolution of the variable $Trading_{b,t}$ in the month of the PSPP announcement and in the following 3 months, for banks with strictly positive holdings of MMA-eligible securities in the preannouncement month. We plot the median (black dot), the 25th/75th percentiles (lower/upper ends of box), and the lower/upper adjacent values (lower/upper limits on whiskers). The lower (upper) adjacent value is the furthest observation within one and a half interquartile ranges of the lower (upper) end of the box.

Table 7
Capital gains versus trading

	2019 announcement		2015 announcement	
	(1)	(2)	(3)	(4)
$[2019m9] \times [(Trading$	-0.135	-0.267	$[2015m1] \times [(Trading$	0.067
$2019m9)/Assets]_b$	[0.184]	[0.188]	$2015m1)/Assets]_b$	[0.087]
$[2019m9] \times [(Capital\ gains$	19.667**		$[2015m1] \times [(Capital\ gains$	44.500***
$2019m9)/Assets]_b$	[9.181]		$2015m1)/Assets]_b$	[7.823]
$[2019m9] \times [(Unrealized$		22.031**	$[2015m1] \times [(Unrealized$	51.837***
$capital\ gains\ 2019m9)/$		[10.954]	$capital\ gains\ 2015m1)/$	[8.042]
$Assets]_b$			$Assets]_b$	
Bank FEs	Yes	Yes	Yes	Yes
Firm-time FEs	Yes	Yes	Yes	Yes
Size, ECB lending, and reserves	Yes	Yes	Yes	Yes
Observations	8,346,925	8,346,925	5,867,308	5,867,308
R-squared	.369	.369	.394	.394

The dependent variable is the change in the logarithm of disbursed loans from bank b to firm f in month t (in percentages). *Capital gains* are measured as the product of lagged quantities of marked-to-market PSPP-eligible securities and the change in price, and they are scaled by assets in the preannouncement month (in percentages). *Trading* is measured as the net purchases of PSPP-eligible securities in the AFS and trading book portfolios, scaled by assets in the preannouncement month (in percentages). *Unrealized capital gains* are the *capital gains* for the securities that are not sold; that is, for each security s , if the bank had a long position, we include the capital gains on the amount that is not sold, and if the bank had a short position, we include the capital gains on the part of the position that remains open; for positions that switch from long to short or vice versa, we assign zero capital gains. Standard errors are clustered at the bank level. * $p < .1$; ** $p < .05$; *** $p < .01$.

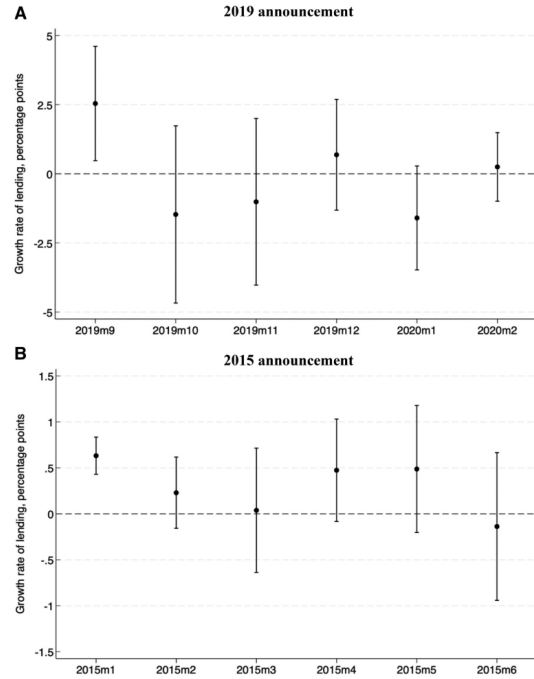


Figure 8

The effects of capital gains on lending

The figure presents the results of a regression for each announcement, in which the dependent variable is the change in the logarithm of disbursed loans from bank b to firm f in month t (in percentages) in the 12-month window around the announcement. Each panel plots the coefficients of $[capital\ gains]_{b,t} / [assets]_{b,t-1}$ interacted with an indicator equal to one in month t , for each post-announcement month. For each month t , the coefficient represents the effect on lending (in percentage points) of increasing capital gains in month t by one standard deviation. The regressions include bank fixed effects, firm-time fixed effects, and the baseline controls (i.e., size, ECB lending, and reserves). Standard errors are clustered at the bank level. The bars represent the 90% confidence interval around each point estimate.

Table 8
The allocation of credit
2019 PSPP announcement

Variables	(1)	(2)	(3)	(4)	(5)	(6)
[2019m9] $\times Q E_b$	0.199** [0.093]	0.226** [0.102]	0.258** [0.107]	0.235** [0.105]		
[2019m9] $\times Q E_b \times [\text{Risky firm}]$		-0.189* [0.100]	-0.209** [0.100]	-0.204** [0.096]	-0.164* [0.092]	-0.153* [0.089]
[2019m9] $\times Q E_b \times [\text{Big firm}]$			-0.217* [0.125]		-0.196 [0.138]	
[2019m9] $\times Q E_b \times [\text{Risky firm}]$ $\times [\text{Tier 1 ratio}]$				0.014** [0.006]		0.009 [0.006]
Observations	4,911,812	4,911,812	4,911,812	4,911,812	4,911,812	4,911,812
R-squared	.348	.348	.348	.348	.351	.351
Bank FEs	Yes	Yes	Yes	Yes		
Bank-time FEs					Yes	Yes
Firm-time FEs	Yes	Yes	Yes	Yes	Yes	Yes
Size, ECB lending, reserves	Yes	Yes	Yes	Yes	Yes	Yes

2015 PSPP announcement

Variables	(1)	(2)	(3)	(4)	(5)	(6)
[2015m1] $\times Q E_b$	0.523*** [0.108]	0.733*** [0.129]	0.771*** [0.109]	1.236** [0.493]		
[2015m1] $\times Q E_b \times [\text{Risky firm}]$		-0.597*** [0.191]	-0.624*** [0.172]	-1.635*** [0.496]	-0.630*** [0.164]	-1.636*** [0.521]
[2015m1] $\times Q E_b \times [\text{Big firm}]$			-0.489 [0.772]		-0.401 [0.819]	
[2015m1] $\times Q E_b \times [\text{Risky firm}]$ $\times [\text{Tier 1 ratio}]$				0.214** [0.091]		0.215** [0.098]
Observations	2,855,099	2,855,099	2,855,099	2,855,099	2,855,099	2,855,099
R-squared	.378	.378	.378	.378	.380	.380
Bank FEs	Yes	Yes	Yes	Yes		
Bank-time FEs					Yes	Yes
Firm-time FEs	Yes	Yes	Yes	Yes	Yes	Yes
Size, ECB lending, reserves	Yes	Yes	Yes	Yes	Yes	Yes

The dependent variable is the change in the logarithm of disbursed loans from bank b to firm f in month t (in percentages). $Q E_b$ denotes the holdings of PSPP-eligible securities valued at MMA, relative to total assets (in percentages). *Risky firm* is a dummy equal to 1 for firms with high credit risk, and *Big firm* is a dummy equal to 1 for firms with revenues exceeding €50 million; both firm-level variables are as of the end of the previous calendar year. The *Tier 1 ratio* is the ratio of Tier 1 capital to risk-weighted assets (in percentages), centered on the first quartile value of its distribution. All regressions include lower-order interaction terms. Standard errors are clustered at the bank level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Economic message. Rebalancing exists, but recapitalization is the dominant lending channel. The lending response is essentially transmitted through capital gains on MMA assets.

5.8 Table 8: who gets the extra credit?

Read:

- In 2019, a median-exposure bank raises lending by about **0.71 percentage points** to nonrisky firms but only about **0.12 percentage points** to risky firms.
- The same exposed banks tilt toward smaller firms as well.
- The quadruple interaction with Tier 1 capital implies that undercapitalized banks derisk more: for a bank at the 25th percentile of the Tier 1 ratio, loan growth to nonrisky firms exceeds that to risky firms by about **0.64 percentage points**; at the 75th percentile, the gap is about **0.37 percentage points**.
- The 2015 results show the same qualitative pattern.

Economic message. The paper is not telling a story of indiscriminate risk taking. The extra lending is safer, especially among weaker-capitalized banks.

5.9 Figure 9 and Table 9: the mechanism generalizes beyond PSPP

Read:

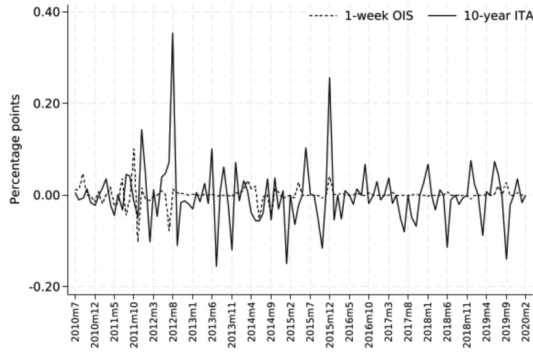


Figure 9
Monetary policy shocks to short- and long-term rates, 2010–2020
The figure plots the monetary policy shocks from the Euro Area Monetary Policy Event-Study Database (EA-MPD, Altavilla et al. 2019) for the period July 2010 to February 2020. These shocks are extracted from movements in market yields in a narrow window around ECB’s regular release of monetary policy information after its meetings. The dotted line represents the shocks to the 1-week overnight index swap rate, and the solid line represents the shocks to 10-year Italian bonds.

Table 9
Other monetary policy shocks (July 2010 to February 2020)

	(1) $\Delta \log L_{b,f,t}$	(2) $\Delta \log L_{b,f,t}$	(3) $\Delta \log L_{b,f,t}$	(4) $\Delta \log L_{b,f,t}$	(5) $\Delta \log L_{b,f,t}$
$[IT10]_t$	-1.170***	-1.215***	-0.909**		-1.296***
$\times [\text{Sovereign MMA (5+)} / \text{Assets}]_{b,t-1}$	[0.339]	[0.358]	[0.395]		[0.488]
$[IT2]_t$		0.163			
$\times [\text{Sovereign MMA (<5)} / \text{Assets}]_{b,t-1}$		[0.259]			
$[IT10]_t$				0.138	
$\times [\text{Sovereign AFS (5+)} / \text{Assets}]_{b,t-1}$				[0.128]	
$[IT10]_t \times [\text{Sovereign AFS (5+)} / \text{Assets}]_{b,t-1}$				-0.985**	
$\times [\text{post-January 2018}]$				[0.487]	
$[IT10]_t$				-1.346***	
$\times [\text{Sovereign trading book (5+)} / \text{Assets}]_{b,t-1}$				[0.423]	
$[IT10]_t$				0.360	
$\times [\text{Sovereign held to maturity (5+)} / \text{Assets}]_{b,t-1}$				[0.290]	
$[IT10]_t \times [\text{Sovereign MMA (5+)} / \text{Assets}]_{b,t-1}$					0.032***
$\times [\text{Tier 1 ratio}]_{b,\text{year}-1}$					[0.011]
$[IT10]_t \times [\text{Sovereign MMA (5+)} / \text{Assets}]_{b,t-1}$					-0.054
$\times [\text{Liquidity} / \text{Assets}]_{b,t-1}$					[0.209]
$[IT10]_t \times [\text{Sovereign MMA (5+)} / \text{Assets}]_{b,t-1}$					0.001
$\times [\text{Tier 1 ratio}]_{b,\text{year}-1}$					[0.009]
$\times [\text{Liquidity} / \text{Assets}]_{b,t-1}$					
Bank FEs	Yes	Yes	Yes	Yes	Yes
Firm-time FEs	Yes	Yes	Yes	Yes	Yes
Size, ECB lending, and reserves	Yes	Yes	Yes	Yes	Yes
Observations	51,461,043	51,461,043	50,575,406	51,461,043	51,461,043
R-squared	.402	.402	.402	.402	.402

The dependent variable is the change in the logarithm of disbursed loans from bank b to firm f in month t (in percentages). The sample period goes from July 2010 to February 2020 and excludes January 2018 (i.e., the month in which the macroprudential accounting rule switched to MMA for available-for-sale securities). $IT10$ is the monetary policy shock for 10-year Italian bonds (in percentage points) from the EA-MPD (Altavilla et al. 2019). *Sovereign MMA (5+)* is the amount of mark-to-market sovereign securities with maturity of at least 5 years, and *Sovereign MMA (<5)* is the amount of sovereign securities with maturity of less than 5 years. *Sovereign AFS (5+)* is the amount of sovereign securities with maturity of at least 5 years in the available-for-sale portfolio; *Sovereign trading book (5+)* and *Sovereign held to maturity (5+)* are defined similarly. *Post-January 2018* is an indicator equal to 1 after January 2018, and zero otherwise. The *Tier 1 ratio* is the ratio of Tier 1 capital to risk-weighted assets at the end of the previous year (in percentages), and *Liquidity* is the sum of central bank deposits and cash at the end of the previous month. All variables that are defined as a fraction of total assets are in percentages. The Tier 1 ratio and liquidity/assets are centered on their respective first quartile levels. All regressions include the lower-order terms unless they are absorbed by the fixed effects. Standard errors are clustered at the bank level. * $p < .1$; ** $p < .05$; *** $p < .01$.

- Figure 9 shows that short-rate shocks are small in the zero-lower-bound years, but long-rate path shocks are often large.
- Table 9 shows that long-rate easing increases lending only through long-duration MMA sovereign exposure.
- A 1-percentage-point negative path-rate shock raises lending by about **0.41 percentage points** for a median-exposure bank at the 25th percentile of Tier 1 capital, but only about **0.29 percentage points** at the 75th percentile.
- The interaction with AFS holdings is significant only after January 2018, when AFS sovereigns move from HCA to MMA.

Economic message. The accounting result is not special to the PSPP. It is a broader statement about how long-rate monetary policy transmits through bank capital.

5.10 Section 7 discussion: policy design

Read:

- The paper writes down a simple capital-constraint framework to show why HCA blocks pass-through and MMA allows it.
- It then proposes a **hybrid accounting rule** that lets changes in the risk-free rate pass through while holding the sovereign spread fixed at its historical value.
- It also shows that HCA plus a static capital requirement is equivalent to MMA plus a time-

varying tightening of capital requirements when yields fall.

Economic message. The paper is ultimately about policy design: how to preserve the stabilizing benefits of HCA without sacrificing monetary-policy transmission.

6 Short summary

- The paper studies two ECB PSPP announcements, 2015 and 2019, using Italian credit-register and supervisory data.
- It shows that QE expands lending only through banks' sovereign holdings that are valued at MMA for regulatory purposes.
- Broad exposure measures that ignore accounting treatment miss the transmission channel.
- The aggregate lending effect is tiny in 2015 and large in 2019 because the share of eligible securities under MMA is much larger in 2019.
- Loan rates fall for exposed banks, confirming a supply shift.
- Capital gains on MMA securities explain the lending response; trading activity does not.
- The extra lending goes mainly to safer firms, and undercapitalized banks derisk the most.
- Similar results appear for broader long-rate monetary shocks, not just the PSPP itself.
- The policy lesson is that macroprudential accounting rules directly shape the effectiveness of monetary policy.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that macroprudential accounting treatment is a first-order determinant of the bank lending channel of QE. Sovereign holdings matter for loan supply only when their price changes pass through to regulatory capital.

7.2 Economic intuition

The intuition is simple. QE raises the price of long-duration sovereigns. If regulation counts those gains in bank capital, constrained banks can lend more. If regulation hides those gains behind HCA, the same monetary policy shock becomes much weaker in real credit markets.

7.3 Takeaway

This is not just a paper about Italy or about one ECB program. It is a paper about the interaction of accounting, prudential regulation, and monetary policy. The central lesson is that stabilizing balance sheets with HCA can come at the cost of muting the pass-through of long-rate monetary policy. The policy frontier is therefore not just “more HCA” or “more MMA,” but designing a rule that protects banks from sovereign-stress volatility without blocking monetary transmission.